DECENTRALIZED EVENT PROTOCOL

TIKETA PROTOCOL

Next-Generation On-Chain Event Ticketing on Botchain

Tiketa is an autonomous Web3 event protocol replacing predatory ticketing monopolies with immutable ERC-1155 smart contracts, capped supplies, and exact native BOT settlement.

ERC-1155
Multi-Tier Capped Tickets

Bytecode supply enforcement

3s Finality
Fast Botchain Settlement

Sub-cent transaction fees

100% P2P
True Self-Custodial Assets

Secondary resale & transfers enabled

The Problem: Legacy Ticketing is Broken

INDUSTRY CRISIS

The $75B global live events ticketing market suffers from systemic monopoly abuses, fraud, and extractive gatekeepers.

1. Rampant Fraud & Counterfeiting

- Over $15B lost annually to fake tickets and scams.
- Duplicate screenshot barcodes scanned at venue gates.
- Fans arrive at sold-out stadiums only to be turned away.
- Zero cryptographic provenance on legacy paper or static apps.

2. Extortionate Hidden Middleman Fees

- Gatekeepers (Ticketmaster, Eventbrite) charge 15% - 35%.
- Hidden convenience, processing, and facility surcharges.
- Siphons immense revenue away from artists and venues.
- Zero transparency into actual organizer pricing.

3. Predatory Bot Scalping

- Automated sniper bots scoop inventory within milliseconds.
- Tickets flipped on secondary markets at 300% - 1,000% markups.
- Creators and performers receive 0% of secondary profits.
- True fans priced out of concerts, sports, and live shows.

4. Zero True Ownership & Arbitrary Lock-In

- Users hold non-custodial entries in proprietary databases.
- Platforms arbitrarily lock, revoke, or cancel transfers.
- Complex walled gardens prevent peer-to-peer flexibility.
- Fans cannot gift or trade tickets without platform consent.

The Solution: The Tiketa Protocol

TRUSTLESS PRIMITIVES

Tiketa replaces extractive middlemen with transparent, decentralized smart contracts deployed on Botchain.

Cryptographic Verifiability (ERC-1155)

- Each ticket tier is an on-chain token ID with capped supply.
- Hard cap enforced at bytecode level (totalSupply + amt <= max).
- Counterfeiting and screenshot duplication are impossible.
- Permanent on-chain event logs for every ticket mint & transfer.

Exact Native BOT Settlement

- Built-in exact payment validation: msg.value == price * amount.
- Rejects underpayment and overpayment automatically.
- Native BOT settlement on Botchain eliminates currency friction.
- Protected against reentrancy attacks via ReentrancyGuard.

Liquid Peer-to-Peer Ownership

- Tickets stay standard ERC-1155 tokens in user self-custody.
- Full support for safeTransferFrom and safeBatchTransferFrom.
- Resale and gifting remain fully open without platform freeze.
- Secondary market freedom without centralized account bans.

Direct Organizer Sovereignty

- Organizers create tiers, set prices, and pause/activate sales.
- Dynamic on-chain price adjustments in real time.
- Instant revenue withdrawal directly to organizer wallet.
- Zero platform approval required to claim ticket sales.

Technical Architecture & Monorepo Stack

PRODUCTION GRADE

Smart Contracts

- Solidity ^0.8.24
- Foundry Framework (forge, cast)
- OpenZeppelin Contracts v5
- ERC1155Supply & Ownable
- ReentrancyGuard Protection
- CEI Security Pattern
- Typed Custom Errors
- 25 / 25 Passing Foundry Tests

Frontend Web dApp

- React 18 + TypeScript
- Vite Next-Gen Bundler
- Tailwind CSS Glassmorphism
- Ethers.js v6 Web3 Library
- Auto Network Add / Switch
- Real-time Explorer Linking
- Dynamic Contract Address Override
- Zero Heavy Dependencies

Botchain Infrastructure

- EVM-Compatible L1 Testnet
- Chain ID: 968 (Hex: 0x3c8)
- Native Token: BOT (18 dec)
- RPC: https://rpc.bohr.life
- Explorer: scan.bohr.life
- ~3-Second Confirmation
- Sub-Cent Gas Execution
- 100% On-Chain Settlement

DEPLOYED PROTOCOL VERIFICATION & INVARIANTS

Contract Address: 0x917524fccd12d61a36918506d5e80679699e534f (Botchain Chain 968)

buy(id, amount): Checks totalSupply(id) + amount <= maxSupply & msg.value == price * amount

withdraw(): Transfers address(this).balance to owner() using nonReentrant & Checks-Effects-Interactions

safeTransferFrom(): Native ERC-1155 peer-to-peer transferability for ticket secondary resale

User Experience & Core dApp Features

LIVE DAPP INTERFACE

Attendee Experience

Interactive Ticket Marketplace:

- Tier cards displaying tier ID, name, and live price in BOT.
- Visual progress bars showing remaining vs. max supply.
- Sold-out & sales paused badges updated live.

Streamlined Minting Flow:

- Quantity selectors with real-time total BOT calculation.
- Auto-prompts wallet to add & switch to Botchain testnet.
- Modal transaction alerts with direct links to Bohr explorer.

My Tickets Portfolio:

- Live query of all owned ERC-1155 token IDs and balances.
- Peer-to-peer ticket transfer modal to send tickets to any wallet.
- Resale-ready standard tokens in self-custody.

Organizer Dashboard

Role-Guarded Administration:

- Access restricted to contract owner (isOwner check).
- Displays accumulated BOT proceeds held in the treasury.
- Single-click revenue withdrawal directly to organizer wallet.

Dynamic Tier Configuration:

- Create new ticket tiers (ID, name, max supply, price in BOT).
- Instant on-chain event emission (TierCreated).
- No redeployment required to add tiers or events.

Live Market Controls:

- Inline price update form (setPrice) to adapt to market demand.
- One-click sales toggle (setTierActive) to pause or resume sales.
- Complete inventory management and supply visibility.

Live Traction & On-Chain Validation

TESTNET DEPLOYED

25 / 25
Foundry Tests Passed
Unit & fuzz test suite

25
Tickets Minted
On-chain tier 1 mints

24
Contract Buy Txs
Direct contract purchases

100%
Tx Success Rate
All confirmed on Botchain

VERIFIED BOTCHAIN TESTNET METRICS

Smart Contract Address	0x917524fccd12d61a36918506d5e80679699e534f	**Verified on Bohr Explorer**
Tier 1: General Admission	Cap: 500 | Minted: 25 | Remaining: 475 | Price: 0.01 BOT	**Active & Selling**
Tier 2: VIP Experience	Cap: 50 | Minted: 0 | Remaining: 50 | Price: 0.05 BOT	**Active & Selling**
Contract Treasury Balance	0.25 BOT accumulated from on-chain ticket sales	**Ready for withdrawal**
Wallet 1 (0xf391...0801)	8 Tier-1 tickets owned | Nonce: 16 | Balance: 0.2075 BOT	**8 Buys Confirmed**
Wallet 2 (0x6D6b...5C96)	8 Tier-1 tickets owned | Nonce: 16 | Balance: 0.2075 BOT	**8 Buys Confirmed**
Wallet 3 (0xE13c...626C)	8 Tier-1 tickets owned | Nonce: 16 | Balance: 0.2075 BOT	**8 Buys Confirmed**

Market Opportunity & Business Model

SUSTAINABLE UNIT ECONOMICS

$85.3B

Global Ticketing TAM by 2028

Expanding at 6.5% annual CAGR

$12B+

Web3 Ticketing Market by 2030

Rapid migration to smart contract passes

90%+

Fee Savings vs. Monopolies

Lower fees for creators and fans

Three Robust Revenue Streams

1. Primary Mint Protocol Fee (0.5% - 1.5%):
- Micro-fee charged on primary ticket mints; provides massive cost savings over Ticketmaster’s 20%+ cut.
- Generates consistent protocol treasury inflow directly correlated with event volume.

2. Secondary Resale Royalties (EIP-2981):
- Configurable 2.5% to 5.0% royalty on secondary ticket resales split between protocol and creators.
- Turns secondary market scalping into a perpetual funding stream for event organizers.

3. Enterprise White-Label & Venue Solutions:
- Dedicated white-label event portal deployments and custom smart contract factories.
- Turnkey venue access control APIs, real-time demand analytics, and sponsor marketing portals.

Competitive Advantage & Differentiation

UNRIVALED POSITIONING

Feature Dimension	Ticketmaster / Eventbrite	YellowHeart / SeatGeek	TIKETA (BOTCHAIN)
Transaction Fee	15% - 35% (Extortionate)	5% - 10%	**< 1.5% (Disruptive)**
Settlement Speed	Days to Weeks (Delayed)	Minutes	**~3 Seconds (Instant)**
Asset Custody	Centralized Database	Semi-Custodial	**100% Self-Custodial (Web3)**
Supply Transparency	Zero (Opaque Black Box)	Partial	**100% Bytecode Enforced**
Gas Cost per Mint	N/A (Hidden Platform Fees)	High ($2 - $15 on ETH)	**Sub-Cent (< $0.001 on Botchain)**
Secondary Resale Rights	Strictly Banned / Locked	Proprietary App Only	**Open ERC-1155 Standard**
Organizer Royalties	0% on Resales	Limited	**Automated On-Chain Payouts**
Code Transparency	Closed Source	Proprietary	**100% Auditable Open Source**

Strategic Roadmap & Milestone Execution

EXECUTION PLAN

Phase 1: Foundation

[COMPLETED - Q4 2026]

- Core ERC-1155 smart contract.
- Supply cap & exact payment rules.
- OpenZeppelin v5 security modules.
- Foundry test suite (25/25 passed).
- Deploy script with cast keystore.
- Botchain testnet deployment.
- Full React 18 + Vite + Tailwind dApp.
- 24 on-chain buy txs verified live.

Phase 2: Scale

[IN PROGRESS - Q1 2027]

- EIP-2981 secondary royalty payouts.
- Decentralized metadata on IPFS.
- Multi-event factory smart contract.
- Venue entrance scanner web app.
- Advanced organizer sales analytics.
- Smart contract security audits.
- Botchain Mainnet deployment.
- First live conference pilot.

Phase 3: Adoption

[PLANNED - Q2 2027]

- Fiat credit card on-ramp to mints.
- Progressive Web App (PWA) with alerts.
- POAP soulbound attendance badges.
- Ticket staking & loyalty perks.
- Secondary marketplace liquidity pool.
- Decentralized organizer DAO.
- Global festival partnerships.
- SDK for third-party venue apps.

Tiketa is Live, Tested, and Proven on Botchain.

By replacing legacy ticketing gatekeepers with mathematical smart contract guarantees, Tiketa empowers event organizers and fans with unprecedented fairness, zero fraud, sub-cent transaction fees, and genuine digital asset ownership.

ON-CHAIN DETAILS & REPOSITORY LINKS

Verified Smart Contract:	0x917524fccd12d61a36918506d5e80679699e534f
Network:	Botchain Testnet (Chain ID: 968 | Native Token: BOT | RPC: https://rpc.bohr.life)
Block Explorer:	https://scan.bohr.life/address/0x917524fccd12d61a36918506d5e80679699e534f
Code Architecture:	Foundry Smart Contracts (Solidity 0.8.24) + React 18 / Vite / Tailwind Web App
Test Verification:	**25 / 25 Passing Foundry Unit & Fuzz Tests | 24 Live Contract Buys Confirmed**